

Level 2 - Java Programs

This document contains three Java programs demonstrating arrays, conditions, loops, and basic data processing.

Program 1

Program 1: Bonus Calculation for Employees

```
import java.util.Scanner;

public class BonusCalculation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double[] salary = new double[10];
        double[] years = new double[10];
        double[] bonus = new double[10];
        double[] newSalary = new double[10];

        double totalBonus = 0;
        double totalOldSalary = 0;
        double totalNewSalary = 0;

        for (int i = 0; i < 10; i++) {
            System.out.print("Enter salary: ");
            salary[i] = sc.nextDouble();

            System.out.print("Enter years of service: ");
            years[i] = sc.nextDouble();

            if (salary[i] <= 0 || years[i] < 0) {
                i--;
                continue;
            }

            for (int i = 0; i < 10; i++) {
                if (years[i] > 5) {
                    bonus[i] = salary[i] * 0.05;
                } else {
                    bonus[i] = salary[i] * 0.02;
                }

                newSalary[i] = salary[i] + bonus[i];

                totalBonus += bonus[i];
                totalOldSalary += salary[i];
                totalNewSalary += newSalary[i];
            }

            System.out.println("Total Bonus: " + totalBonus);
            System.out.println("Total Old Salary: " + totalOldSalary);
            System.out.println("Total New Salary: " + totalNewSalary);
        }
    }
}
```

Program 2

Program 2: Friends Information Analysis

```
import java.util.Scanner;

public class FriendsInfo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String[] names = {"Amar", "Akbar", "Anthony"};
    }
}
```

```

int[] age = new int[3];
double[] height = new double[3];

for (int i = 0; i < 3; i++) {
    System.out.print("Enter age of " + names[i] + ": ");
    age[i] = sc.nextInt();

    System.out.print("Enter height of " + names[i] + ": ");
    height[i] = sc.nextDouble();
}

int youngestIndex = 0;
int tallestIndex = 0;

for (int i = 1; i < 3; i++) {
    if (age[i] < age[youngestIndex]) {
        youngestIndex = i;
    }

    if (height[i] > height[tallestIndex]) {
        tallestIndex = i;
    }
}

System.out.println("Youngest friend: " + names[youngestIndex]);
System.out.println("Tallest friend: " + names[tallestIndex]);
}
}

```

Program 3

Program 3: Reverse Digits of a Number

```

import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();
        int num2 = num;

        if(num < 0){
            num *= -1;
        }

        int num3 = num;
        int num4 = 0;
        int index = 0;

        while(num != 0){
            num /= 10;
            index++;
        }

        int[] arr = new int[index];

        for(int i = 0; i < index; i++){
            arr[i] += num3 % 10;
            num3 /= 10;
        }

        if(num2 > 0){
            for(int i : arr){
                System.out.print(i);
            }
        }
        else{
            System.out.print("-");
            for(int j : arr){
                System.out.print(j);
            }
        }
    }
}

```